

In vitro effect of herbal products against *Histomonas meleagridis*

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Abstract

The ban on chemotherapeutics against *Histomonas meleagridis* in the European Union has left turkey producers without an effective treatment against histomonosis. It has encouraged the development of alternative control methods amongst which are a number of herbal products. In the present study the *in vitro* effect of four herbal products against *H. meleagridis* was examined. After suspension of the herbal products in Dwyer's culture medium used for subculturing of a viable culture, parasite growth was monitored by frequently counting the histomonads until 72 h of incubation. The solid products AromabioticTM, EnteroguardTM, and Protophyt SPTM as well as the positive control products dimetridazole and Histostat-50TM, were tested in twofold serial dilutions ranging in concentration from 400 to 12.5 µg/mL, while the liquid product Protophyt BTM was tested in concentrations ranging from 0.24% to 0.008% (v/v). The herbal products EnteroguardTM and Protophyt BTM as well as dimetridazole and Histostat-50TM, two chemotherapeutics with known antihistomonal activity, inhibited the growth of *H. meleagridis*, while no antihistomonal effect was found for AromabioticTM and Protophyt SPTM.

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1. Introduction

Histomonas meleagridis is an amoeboid protozoan with a single flagellum and may cause histomonosis (infectious enterohepatitis; blackhead disease) in galliforms and other species of birds. The disease can result in high mortality in turkeys, while in chicken histomonosis is generally less severe (McDougald, 1997). However, chicken flocks may form a disease reservoir with the cecal worm *Heterakis gallinarum* acting as an intermediate vector between chicken and turkey flocks. The very fragile *H. meleagridis* protozoa

may infect heterakid eggs (Graybill and Smith, 1920) and thus remain infective in soil or faeces for years (McDougald, 2005).

During the last decades several effective chemotherapeutics like nifursol, arsenical compounds (e.g., nitarosone and carbasone), some nitrofurans, and nitroimidazoles (e.g., dimetridazole, ipronidazole, and ronidazole) were available as preventive feed additives or as curative therapeutics (Hegnig et al., 1999; McDougald, 2005). However, increasing concern for the presence of residues in food and possible toxicity and carcinogenicity of these therapeutics led to their ban in Europe some years ago, while in the US at present only nitarosone is permitted (Byrne, 2001; Zenner et al., 2003; McDougald, 2005).

In recent studies several compounds and products were found to have an antihistomonal activity. These included nitroimidazoles, arsenicals (Callait et al., 2002; Hu and McDougald, 2004), volatile oils of

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Cinnamomum aromaticum, *Citrus limon*, and *Allium sativum* (Zenner et al., 2003), a plant-derived product NatustatTM (Duffy et al., 2005), several natural organic compounds (Grabensteiner et al., 2007) and other chemotherapeutics (tinidazole, metronidazole, ornidazole, doloxanide furoate, albendazole, mebendazole, paromomycine sulfate, 5,7-diiodo-8-hydroxy-quinoline, and carbadox) which are active against other protozoan parasites (Hu and McDougald, 2004).

A number of herbal products with claimed antibacterial and/or antiprotozoan activity have been marketed in recent years for addition to poultry feed or drinking water. However, most of these compounds have not been tested for antihistomonal activity. Therefore, the *in vitro* effect of some herbal products against *H. meleagridis* was examined in the present study.

An *in vitro* experiment is a cost-effective method to conveniently screen products for possible antihistomonal activity. An observed *in vitro* effect, however, may be caused by an antibacterial effect rather than a direct effect on the parasite since *H. meleagridis* growth is dependent on the presence of cecal bacteria in the dioxenic culture (McDougald, 2005). Another shortcoming of an *in vitro* experiment is that the efficacy of compounds may be missed in case they only become active in the host, e.g., after metabolization (McDougald, 2005). Therefore, an observed *in vitro* effect should always be confirmed in an *in vivo* experiment.

2. Materials and methods

2.1. Products

Two drugs with known activity against histomonosis were used as control products during the *in vitro* experiments: dimetridazole (1,2-dimethyl-5-nitroimidazole; Sigma, Zwijndrecht, The Netherlands) and Histostat-50TM (Alpharma Inc., Fort Lee, USA) consisting of 50% nitarosone (4-nitrophenylarsonic acid) with calcium carbonate and rice hull as carrier material.

Three herbal products were examined. AromabioticTM (Vitamix, Belgium) is a feed additive (advised dosage 0.3%) derived from tropical plants. It consists of the medium chain fatty acids capronic acid, caprylic acid and capric acid. EnteroguardTM (Orffa, Giessen, The Netherlands) is a feed additive (dosage 0.5–2%) consisting of lyophilized garlic and cinnamon infusion with active compounds allicin and cinnamaldehyde and other thiosulfonates. ProtophytTM (Phytosynthèse, Riom, France) is a product based on oils extracted from leaves of cinnamon (8–12%), garlic (3–5%), rosemary (2–3%) and

lemon (1–2%). Both the feed additive form of this product, Protophyt SPTM (dosage 0.3%), as well as the drinking water form, Protophyt BTM (dosage 0.3–1%), were examined.

2.2. *H. meleagridis* isolate and culture

The *H. meleagridis* strain Turkey/Deventer/NL/AL327-type I/03 (Van der Heijden et al., 2006) was isolated from a turkey presenting typical liver and caecal lesions (Van der Heijden et al., 2005). The isolate, stored at –175 °C in liquid nitrogen in 1 mL aliquots, was resuscitated in 12.5 mL of Dwyer medium (Dwyer, 1970) at 40 °C, and consisting of 85% Medium 199 with Hanks salts (Gibco-BRL), 10% heat-inactivated horse serum (Gibco-BRL), 5% chicken embryo homogenate of 8-day-old eggs and 12 mg of rice powder (Arrowhead Mills, Hereford, USA). After an incubation period of 3 days at 40 °C subculturing was performed by inoculating 1 mL of histomonad culture into five 70 mL tissue culture flasks (Becton Dickinson, France) containing 12.5 mL of modified Dwyer's medium prewarmed at 40 °C and consisting of 90% Medium 199, 10% horse serum and 100 mg (0.8%) of rice powder (Van der Heijden and Landman, 2007). After 24 h of incubation the contents of the five tissue culture flasks were pooled. The concentration of histomonads was determined using a Bürker-Türk haematocytometer and phase contrast microscopy at 200× magnification (minimum required 0.5×10^6 histomonads/mL).

2.3. *In vitro* experiments

Three *in vitro* experiments were performed. In the first experiment all products were tested at 400 µg/mL, or 0.24% (v/v) for Protophyt BTM, in PBST (phosphate-buffered saline with 0.2% Tween 80). In the second experiment, the products that had an inhibitory effect at the high concentration used in the first experiment, were tested serially diluted in PBST. In the third experiment, the products that were not effective in the first experiment, were tested serially diluted in PBS without Tween. In all experiments negative controls (PBST or PBS) and positive controls (dimetridazole at 400 and 12.5 µg/mL) were included. All experiments were repeated.

Fresh stocks of the herbal products were prepared for each experiment by suspending 100 mg in 10 mL of PBST or PBS except for the liquid Protophyt BTM that was diluted 1:10. Suspensions were vortexed three times during 20 s and incubated for 1 h at room temperature.

Then, the suspensions were vortexed again during 20 s. From these stock suspensions twofold serial dilutions were made in PBST or PBS using 1-mL volumes (concentrations in culture: 400, 200, 100, 50, 25, and 12.5 $\mu\text{g/mL}$). In each tissue culture flask, 7 mL of the pooled 24 h *H. meleagridis* culture was added to 5.5 mL of the freshly prepared modified Dwyer's medium prewarmed at 40 °C. Subsequently, 521 μL of the product suspension was added and mixed by gentle swirling. Cultures were incubated at 40 °C and counted at 8, 24, 32, 48, 54, and 72 h (during the first experiment also at 4, 20, 28, and 44 h).

2.4. Statistical analysis

The log-transformed duplicate parasite counts of cultures with product additions were compared to the relevant negative control (PBS or PBST) using Student's *t*-test with a 95% confidence interval.

3. Results

3.1. Experiment I

The histomonad counts for EnteroguardTM, AromabioticTM, Protophyt SPTM, and the negative control PBST remained high throughout the experiment (Fig. 1). Almost identical results were obtained when the experiment was repeated. For Protophyt BTM, dimetridazole, and Histostat-50TM the number of parasites declined gradually with time. For Protophyt BTM the number of histomonads was already significantly lower than for PBST at 4 h of incubation and dropped below the detection limit ($10^{3.7}$ histomonads/mL) after 20 h, while

for dimetridazole and Histostat-50TM the parasite counts for the first time were significantly lower than for PBST after 24 and 28 h of incubation, respectively.

3.2. Experiment II

No histomonads were detected throughout the experiment when 0.24% Protophyt BTM was added to the culture medium (Fig. 2). At a concentration of 0.12% the number of histomonads decreased during the experiment and dropped below the detection limit after 54 h. Some inhibition of *H. meleagridis*' growth was also observed for 0.06% Protophyt BTM. At repetition, the inhibitory effect of 0.24% Protophyt BTM was confirmed. However, at lower concentrations the numbers of histomonads did not decline over time. On average, significantly lower parasite counts were only found for 0.24% Protophyt BTM compared to PBST.

Complete elimination of histomonads from culture was observed after 32 h when dimetridazole was added to the culture medium at concentrations ranging from 50 to 400 $\mu\text{g/mL}$ (Fig. 3). At a concentration of 12.5 $\mu\text{g/mL}$ dimetridazole induced a temporary decrease in histomonad counts until 32 h of incubation. Thereafter, the number of parasites increased again. Upon repetition, a similar effect was found, though now this concentration of 12.5 $\mu\text{g/mL}$ dimetridazole did not fully inhibit growth of histomonads at 32 h. Moreover, a temporary decrease in histomonads was also observed at a concentration of 25 $\mu\text{g/mL}$ dimetridazole. On average, compared to PBST, significantly lower parasite counts for dimetridazole concentrations between 400 and 50 $\mu\text{g/mL}$ were found from 24 h of incubation onwards.

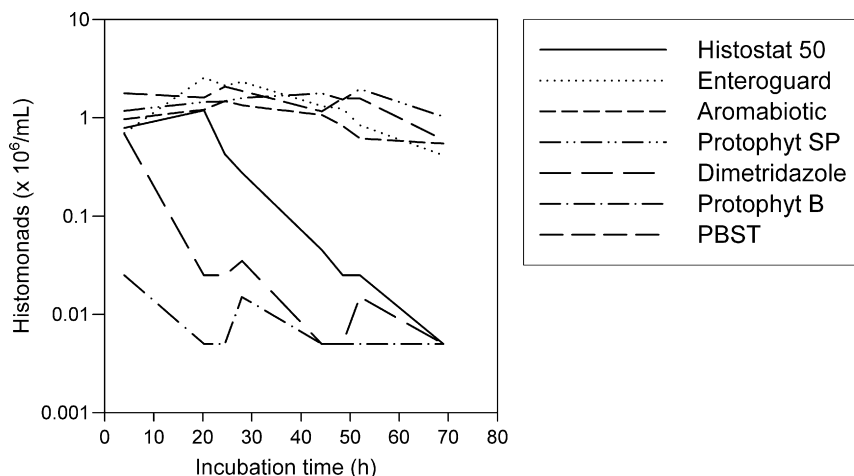


Fig. 1. Histomonads counts of cultures with different products added at a concentration of 400 $\mu\text{g/mL}$ or 0.24% for Protophyt BTM in phosphate-buffered saline with Tween 80 (PBST).

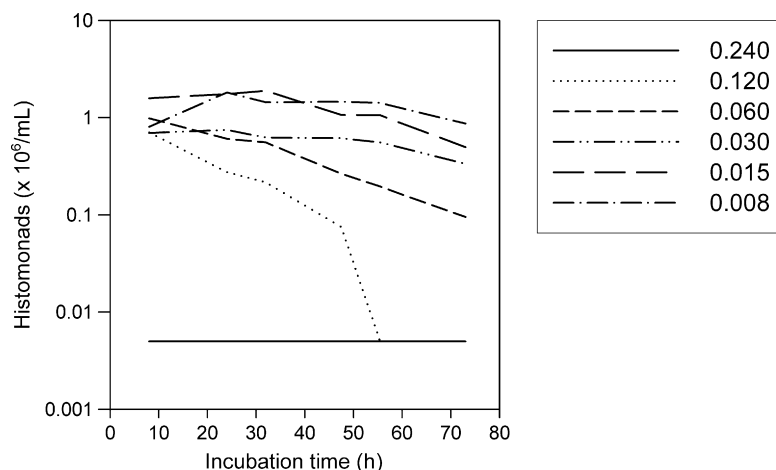


Fig. 2. Histomonads counts of cultures with serial dilutions of Protophyt BTM (%) in PBST.

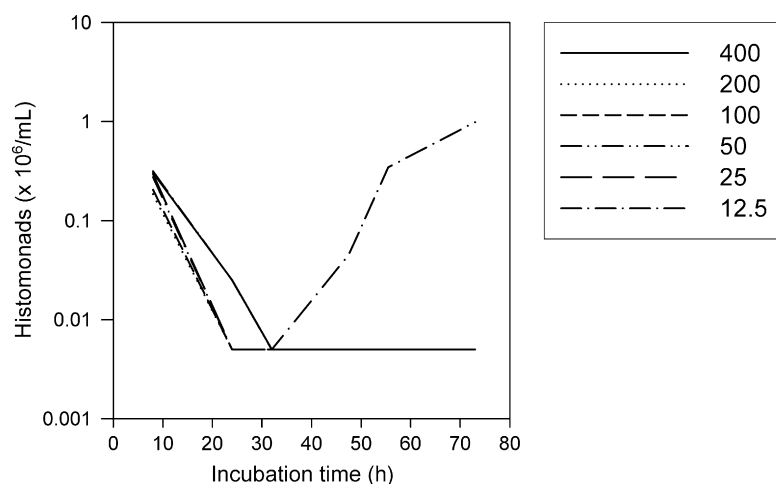


Fig. 3. Histomonads counts of cultures with serial dilutions of dimetridazole (μg/mL) in PBST.

Histostat-50TM yielded significantly lower histomonad counts compared to PBST from 32 h of incubation onwards at concentrations of 400 and 200 μg/mL. A concentration of 50 μg/mL Histostat-50TM (Fig. 4) also showed complete elimination of parasites, while this was not the case at a concentration of 100 μg/mL. Nevertheless, at repetition opposite results were obtained. Histostat-50TM concentrations of 25 or 12.5 μg/mL did not significantly affect parasite growth.

3.3. Experiment III

Three products (AromabioticTM, Protophyt SPTM, and EnteroguardTM), that showed no inhibitory effect on histomonad growth when suspended in PBST

(experiment I), were also tested after suspension and serial dilution in PBS (i.e., without Tween 80).

The numbers of histomonads were comparable throughout the experiment for both AromabioticTM and Protophyt SPTM at all concentrations tested and did not significantly differ from the counts of the negative control PBS.

In the first test (Fig. 5A) with EnteroguardTM at the highest concentration (400 μg/mL) already after 32 h lower numbers of histomonads were found. Regarding the other concentrations tested hardly or only a small (50 μg/mL) decrease in the numbers of parasites was observed toward the end of the first test. In the second test (Fig. 5B) again the highest concentration (400 μg/mL) of EnteroguardTM resulted in a clear decrease of

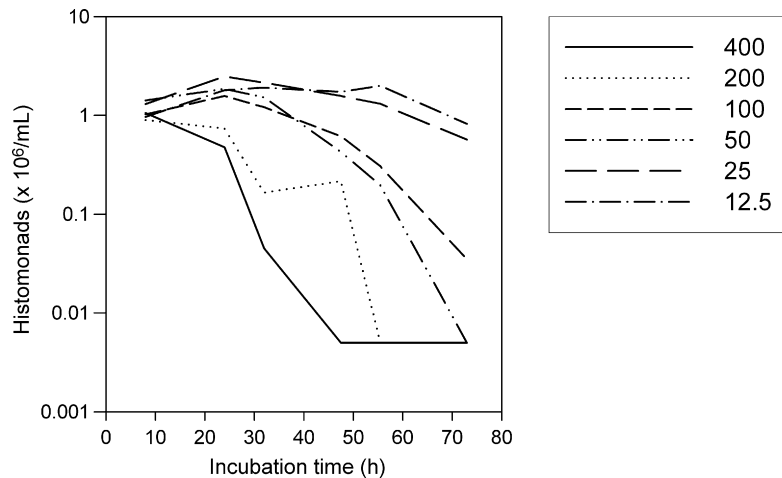


Fig. 4. Histomonads counts of cultures with serial dilutions of Histostat-50TM ($\mu\text{g/mL}$) in PBST.

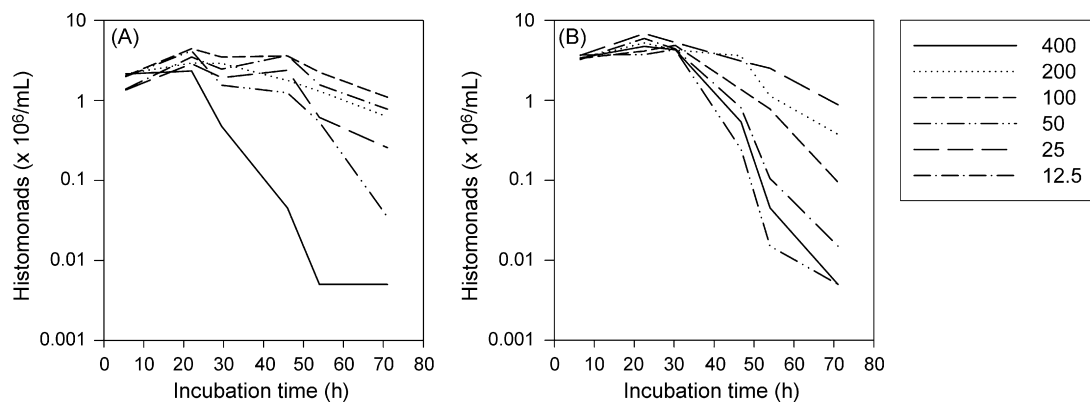


Fig. 5. Histomonads counts of cultures with serial dilutions of EnteroguardTM ($\mu\text{g/mL}$) in PBS: (A) first test; (B) repetition.

the number of histomonads, but after 48 h. Other concentrations resulted in lower counts. The timing and degree of count decrease was not well correlated with the concentration. On average, only the 400 and 200 $\mu\text{g/mL}$ after 72 h of incubation had a significantly lower parasite count in comparison with PBS.

The control groups (dimetridazole in two concentrations as the positive control, or PBS as a negative control in fourfold) that were included in this test showed similar patterns as in previous experiments.

4. Discussion

During the last decade only a few studies have been published in which the activity of products against *H. meleagridis* was examined. Nitroimidazoles (Callait et al., 2002; Hu and McDougald, 2004) and furazolidone (Callait et al., 2002) were found to be effective chemotherapeutics as already reported in the early days

of histomonas research. However, the withdrawal of these products (Byrne, 2001), because of toxic and possible carcinogenic properties, prompted the quest for alternative products with antihistomonal activity. Volatile oils extracted from leaves of *C. aromaticum*, *C. limon* and *A. sativum* were found to have an antihistomonal activity against *H. meleagridis* *in vitro* before (Zenner et al., 2003), and Grabensteiner et al. (2007) found *in vitro* susceptibility of *H. meleagridis* isolates to an essential oil mixture derived from thyme and rosemary, carvacrol, *Caissa* oil and *Quillaja saponaria* saponin.

In the present study, a number of feed additives and a drinking water additive were tested *in vitro*, with dimetridazole as a positive control with known antihistomonal activity. The effective dose of dimetridazole was found to be approximately 12.5 $\mu\text{g/mL}$. This is in agreement with results obtained in other studies (Callait et al., 2002; Hu and McDougald, 2004).

At higher concentrations (400–50 $\mu\text{g/mL}$) the inhibition of growth was complete and irreversible throughout the test period, while partly and temporary inhibition of growth was observed at the lowest concentration tested (12.5 $\mu\text{g/mL}$). Probably a high concentration of dimetridazole is fatal to *H. meleagridis*, whereas at lower concentrations only a histomonostatic effect occurs. These results parallel other *in vitro* (Callait et al., 2002) and *in vivo* studies (McDougald, 2005).

The effective *in vitro* concentration of nitarson (an arsenic compound) is not known. Histostat-50TM in the present study inhibited growth of parasites at concentrations of 50 $\mu\text{g/mL}$ or higher. The exposure time needed to inhibit the growth of *H. meleagridis* was dose-dependent and varied between approximately 30 h (400 $\mu\text{g/mL}$) and more than 60 h (50 or 100 $\mu\text{g/mL}$), which was longer than the exposure time for dimetridazole. This delay may have been caused by the carrier material of Histostat-50TM partly consisting of rice hull. Such materials could stimulate growth of *H. meleagridis* during *in vitro* experiments, because rice powder is known to be an important constituent of the culture medium for *H. meleagridis* (Van der Heijden et al., 2005). Also other flours like corn, starch, rye, oat, barley, and buckwheat are suitable replacements for rice powder (McDougald, 2005).

In the present study EnteroguardTM (with active compounds allicin from garlic and cinnamaldehyde from cinnamon) was effective against *H. meleagridis*. Tween 80 was found to mask the antihistomonal effect, since only inhibition of parasite growth was seen when the product was suspended and serially diluted in PBS without Tween 80. The masking effect was probably not directed against the active compounds since Zenner et al. (2003) found an *in vitro* antihistomonal activity in oils derived from leaves of garlic and cinnamon, and had also used Tween 80 as a detergent in their solvent.

The correlation between EnteroguardTM concentration and the antihistomonal effect was relatively poor and also the reproducibility antihistomonal effect was worse than that of the other products. A possible explanation could be the composition of the product. Non-effective components could serve as a nutrient for histomonads and subsequently mask the action of the antihistomonal compounds. This could lead to the irregular results observed in the present study.

The results of both Protophyt SPTM (feed additive) and Protophyt BTM (drinking water additive) were completely opposite. Protophyt BTM, when tested at the highest concentration, showed a significant effect against *H. meleagridis*, while no antihistomonal effect was seen at all with Protophyt SPTM. Both products are

based on the same ingredients (volatile oils from garlic, cinnamon, rosemary and lemon). The difference might have been caused by differences in concentrations of the active compounds in the culture medium, e.g., due to a better dispersion of the volatile oils in the liquid product facilitated by the addition of emulsifiers.

AromabioticTM is based on medium chain fatty acids like capronic acid. The mode of action against bacteria is based on the assumption that these fatty acids incorporate in the cell wall structure and cause leakage of the cells. In addition, this product causes an intracellular rise in pH caused by dissociation. It is unknown whether the same mechanisms would apply to eukaryotic cells like histomonads. Nevertheless, in the present study no antihistomonal effect of AromabioticTM was found.

Although *in vitro* testing is a practical method to screen products or compounds for a possible antihistomonal effect, it has some drawbacks. The results are not always very reproducible; variation was observed in the time needed for inhibition of parasite growth, as well as in the minimal inhibitory concentrations. However, in a screening test it is more important that candidate products consistently inhibit histomonal growth rather than yielding exact *in vitro* inhibitory concentrations and incubation times. Further, the nature and composition of the solvent used in an *in vitro* study may also influence the results. Most feed additives contain carrier materials (e.g., rice hull) that cannot be dissolved and which may stimulate histomonal growth, while the use of active oily compounds requires the addition of detergents for their dissolution.

In vitro testing of antihistomonal activity may also result in false positive results if the products are bactericidal to the cecal bacteria present in the *H. meleagridis* culture (McDougald, 2005). It is generally accepted that *H. meleagridis* will only grow in dioxenic culture with cecal bacteria or in axenic culture with killed bacteria added to the medium (Lesser, 1960). Indeed, using an antibiotic (streptomycin) complete inhibition of the cecal bacterial growth in the medium was achieved (data not shown). This also resulted in complete inhibition of growth of the *Histomonas* strain used in the present study. Remarkably, Callait et al. (2002) reported successful culture of *H. meleagridis* up to 48 h after specific destruction of the bacterial flora, concluding that the inhibition of histomonads growth in their study was directly caused by the product tested. In the present study not a single product caused a visually less turbid culture, suggesting that the products did not inhibit bacterial growth. However, as the intestinal flora is complex and consists of different species it cannot be

excluded that a product may specifically eliminate one (or several) bacterial species that are essential for the growth of *H. meleagridis*. This is another reason why it is very important to confirm the *in vitro* efficacy of products in an *in vivo* study. Therefore, the observed antihistomonal *in vitro* effects of Protophyt BTM and EnteroguardTM should be confirmed in an animal experiment before final conclusions about their antihistomonal efficacy can be drawn.

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